Law of Sines and Law of Cosines

Section 1: The Law of Sines

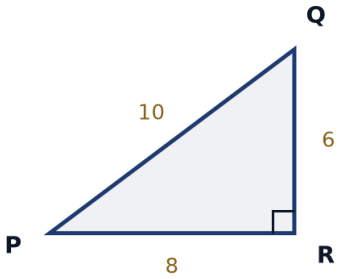
1 State the Law of Sines for any triangle ABC.

a. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

b. $a^2 = b^2 + c^2 - 2bc\cos A$

c. $a\sin A = b\sin B = c\sin C$

d. $\frac{a}{\cos A} = \frac{b}{\cos B}$



2 In $\triangle ABC$, $A = 30^\circ$, $B = 45^\circ$, and $a = 8$. Find the exact length of side b.

a. $8\sqrt{2}$

b. $4\sqrt{2}$

c. $8\sqrt{3}$

d. 16

3 Under which given information cases can the Law of Sines be directly applied to begin solving a triangle?

a. AAS or ASA

b. SAS or SSS

c. SSS only

d. Right triangles only

Section 2: The Ambiguous Case (SSA)

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4 In the ambiguous SSA case with acute angle A, what does $h = b\sin A$ represent?

a. The altitude from vertex C to side c

b. The perimeter of the triangle

c. The hypotenuse

d. The length of side a

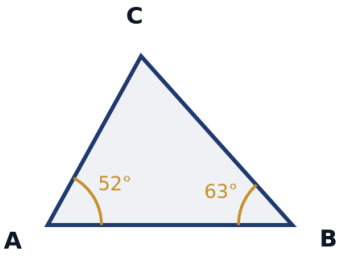
5 If $A = 30^\circ$, $b = 10$, and $a = 4$, how many triangles can be formed?

a. 0 (no triangle exists)

b. 1 triangle

c. 2 triangles

d. Infinitely many



6 If $A = 30^\circ$, $b = 10$, and $a = 7$, how many triangles can be formed?

a. 2 triangles

b. 1 triangle

c. 0 triangles

d. Cannot be determined

Section 3: The Law of Cosines

7 State the Law of Cosines to solve for side c given sides a, b and included angle C.

a. $c^2 = a^2 + b^2 - 2ab\cos C$

b. $c^2 = a^2 + b^2 + 2ab\cos C$

c. $c = a\cos B + b\cos A$

d. $c^2 = a^2 + b^2 - ab\sin C$

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8 In $\triangle ABC$, $a = 5, b = 7$, and $C = 60^\circ$. Find the exact length of side c.

a. $\sqrt{39}$

b. $\sqrt{74}$

c. $\sqrt{49} = 7$

d. 6

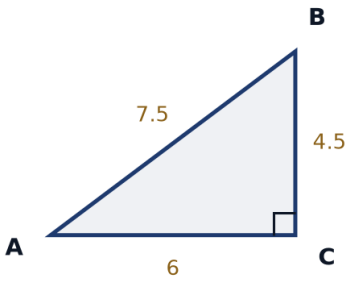
9 In $\triangle ABC$, $a = 3, b = 5, c = 7$. Find the exact value of $\cos C$.

a. $-1/2$

b. $1/2$

c. $-1/5$

d. $-3/7$



Section 4: Area of an Oblique Triangle

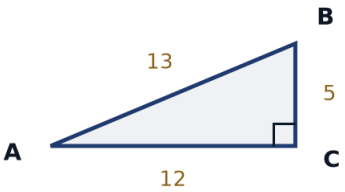
10 State the sine formula for the area of $\triangle ABC$ given sides a and b and angle C.

a. $\text{Area} = \frac{1}{2}ab\sin C$

b. $\text{Area} = ab\sin C$

c. $\text{Area} = \frac{1}{2}ab\cos C$

d. $\text{Area} = \frac{a+b}{2}\sin C$



11 Find the exact area of $\triangle ABC$ if $a = 6, b = 10$, and $C = 30^\circ$.

a. 15

b. 30

c. $15\sqrt{3}$

d. 7.5

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12 Heron's formula states Area $= \sqrt{s(s-a)(s-b)(s-c)}$. Find the area of a triangle with sides 5, 6, and 7.

a. $6\sqrt{6}$

b. $12\sqrt{3}$

c. 18

d. 14

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